

Syllabus of final exam

Group A : (20 number)

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Group B : (30 number)

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Chap 1: partial Differentiation

Defn: In mathematics a partial derivative of a function of several variables is its derivative with respect to one of those variables, with the others held constant. partial derivatives are used in vector calculus and differential geometry.

$$\text{Let } u = f(x, y)$$

$$\therefore \frac{\partial}{\partial x} (u) = u_x ; \text{ when } y \text{ constant}$$

$$\frac{\partial}{\partial y} (u) = u_y ; \text{ when } x \text{ constant}$$

Notation: $u = f(x, y) \rightarrow u_x, f_x, \frac{\partial u}{\partial x}$
 $\rightarrow u_y, f_y, \frac{\partial u}{\partial y}$

[Q1] If $u = \log(x^2 + y^2)$ then show that

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$$

Given that, $u = \log(x^2 + y^2)$ ——— (1)

partially
Now diff. (1) w.r. to x two times,

we get, $\frac{\partial u}{\partial x} = \frac{1}{x^2 + y^2} \cdot 2x$

$$\text{Also, } \frac{\partial^2 u}{\partial x^2} = \frac{(x^2+y^2) \cdot 2 - 2x \cdot 2x}{(x^2+y^2)^2} \quad \left[\frac{d}{dx} u \cdot v \right]$$

$$= \frac{2x^2 + 2y^2 - 4x^2}{(x^2+y^2)^2}$$

$$= \frac{2y^2 - 2x^2}{(x^2+y^2)^2} \quad \text{--- (II)}$$

Again,
partial diff. (I) w.r. to y two times we get,

$$\frac{\partial u}{\partial y} = \frac{1}{x^2+y^2} \cdot 2y$$

$$\text{Also, } \frac{\partial^2 u}{\partial y^2} = \frac{(x^2+y^2) \cdot 2 - 2y \cdot 2y}{(x^2+y^2)^2}$$

$$= \frac{2x^2 + 2y^2 - 4y^2}{(x^2+y^2)^2}$$

$$= \frac{2x^2 - 2y^2}{(x^2+y^2)^2} \quad \text{--- (III)}$$

Now adding (II) & (III) we get,

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = \frac{2y^2 - 2x^2}{(x^2+y^2)^2} + \frac{2x^2 - 2y^2}{(x^2+y^2)^2}$$

$$= \frac{2y^2 - 2x^2 + 2x^2 - 2y^2}{(x^2+y^2)^2}$$

$$= 0$$

Hence, $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$ (shown)

② If $u = \ln(x^3 + y^3 + z^3 - 3xyz)$ then show that, $\left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} + \frac{\partial}{\partial z}\right)^2 u = \frac{-9}{(x+y+z)^2}$

Solⁿ:

Given that,

$$u = \ln(x^3 + y^3 + z^3 - 3xyz) \text{ --- ①}$$

Now partially diff. ① w.r. to x, y, z respectively we get,

$$\begin{aligned} \frac{\partial u}{\partial x} &= \frac{1}{(x^3 + y^3 + z^3 - 3xyz)} \cdot (3x^2 - 3yz) \\ &= \frac{3(x^2 - yz)}{(x^3 + y^3 + z^3 - 3xyz)} \end{aligned}$$

Similarly,

$$\frac{\partial u}{\partial y} = \frac{3(y^2 - xz)}{(x^3 + y^3 + z^3 - 3xyz)}$$

$$\frac{\partial u}{\partial z} = \frac{3(z^2 - xy)}{x^3 + y^3 + z^3 - 3xyz}$$

$$\begin{aligned}
 \text{Now, } \frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} &= \frac{3(x^2 + y^2 + z^2 - xy - yz - zx)}{x^3 + y^3 + z^3 - 3xyz} \\
 &= \frac{3(x^2 + y^2 + z^2 - xy - yz - zx)}{(x+y+z)(x^2 + y^2 + z^2 - xy - yz - zx)} \\
 &= \frac{3}{(x+y+z)}
 \end{aligned}$$

$$\begin{aligned}
 \text{Now, L.S} &= \left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} + \frac{\partial}{\partial z} \right)^2 u \\
 &= \left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} + \frac{\partial}{\partial z} \right) \left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} + \frac{\partial}{\partial z} \right) u \\
 &= \left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} + \frac{\partial}{\partial z} \right) \left(\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} \right) \\
 &= \left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} + \frac{\partial}{\partial z} \right) \frac{3}{(x+y+z)} \\
 &= \frac{\partial}{\partial x} \left(\frac{3}{x+y+z} \right) + \frac{\partial}{\partial y} \left(\frac{3}{x+y+z} \right) + \frac{\partial}{\partial z} \left(\frac{3}{x+y+z} \right) \\
 &= 3 \cdot \frac{-1}{(x+y+z)^2} \cdot 1 + 3 \cdot \frac{-1}{(x+y+z)^2} + 3 \cdot \frac{-1}{(x+y+z)^2} \\
 &= \frac{-9}{(x+y+z)^2} = \text{R.S} \\
 &\quad \text{(Hence, shown)}
 \end{aligned}$$

☐ State and prove Euler's theorem on Homogeneous function,

Homogeneous function: If the sum of the powers of the each term of the function are equal is called Homogeneous function.

If u ~~and~~ v be a homogeneous function of degree n in x, y , we can write,

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = nu$$

Q2 If $u = \tan^{-1} \frac{x^3 + y^3}{x + y}$ then using Euler's

theorem show that, $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = \sin 2u$.

Sol

Given that,

$$u = \tan^{-1} \frac{x^3 + y^3}{x + y}$$

$$\Rightarrow \tan u = \frac{x^3 + y^3}{x + y}$$

$$= \frac{x^3 (1 + y^3/x^3)}{x(1 + y/x)}$$

$$= x^2 F(y/x) \quad \left[\text{where} \right.$$

$$F(y/x) = \frac{1 + y^3/x^3}{1 + y/x}$$

Now by Euler's theorem we have,

$$x \frac{\partial}{\partial x} (\tan u) + y \frac{\partial}{\partial y} (\tan u) = 2 \tan u$$

$$\Rightarrow x \cdot \sec^2 u \cdot \frac{\partial u}{\partial x} + y \cdot \sec^2 u \frac{\partial u}{\partial y} = 2 \tan u$$

$$\Rightarrow x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = 2 \cdot \frac{\sin u}{\cos u} \cdot \cos^2 u$$

$$= 2 \sin u \cdot \cos u$$

$$= \sin 2u$$

$$\text{Hence, } x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = \sin 2u. \text{ (shown)}$$

Chapter 2: Maxima and Minima

Extreme Value

Condition:

- (i) The function $f(x)$ will be maximum at $x = a$ if $f'(a) = 0$ and $f''(a) < 0$
- (ii) The function $f(x)$ will be minimum at $x = a$ if $f'(a) = 0$ and $f''(a) > 0$.
- (iii) The function $f(x)$ has neither maximum nor minimum at $x = a$ if $f'(a) \neq 0$.

TYPE 1

Q1 find the maximum and minimum value of $f(x) = 2x^3 - 9x^2 + 12x - 3$.

Given that,

$$f(x) = 2x^3 - 9x^2 + 12x - 3$$

$$\therefore f'(x) = 6x^2 - 18x + 12$$

also,

$$f''(x) = 12x - 18$$

Now for, maximum and minimum values,

$$f'(x) = 0$$

$$\Rightarrow 6x^2 - 18x + 12 = 0$$

$$\Rightarrow x^2 - 3x + 2 = 0$$

$$\Rightarrow x^2 - 2x - x + 2 = 0$$

$$\Rightarrow x(x-2) - 1(x-2) = 0$$

$$\Rightarrow (x-1)(x-2) = 0$$

$$\therefore x = 1, 2.$$

Now when $x = 1$, then $f''(x) = 12 \cdot 1 - 18$

$$= -6 < 0.$$

So at $x = 1$ the given function will be maximum.

Hence the maximum value $= 2 \cdot 1^3 - 9 \cdot 1^2 + 12 \cdot 1 - 3$

$$= 2 \text{ Ans}$$

Again,

When $x = 2$, then $f''(x) = 12 \cdot 2 - 18$

$$= 6 > 0$$

So at $x=2$ the given function will be minimum.

$$\text{Hence the minimum value} = 2 \cdot 2^3 - 9 \cdot 2^2 + 12 \cdot 2 - 3$$

$$= 1 \quad \text{Ans}$$

(2) Show that the maximum value of $x + \frac{1}{x}$ is less than its minimum value.

Given that,

$$f(x) = x + \frac{1}{x} = x + x^{-1} \quad \text{--- (1)}$$

$$\therefore f'(x) = 1 - \frac{1}{x^2}$$

Also,

$$f''(x) = \frac{2}{x^3}$$

$$\text{Let } f'(x) = 0$$

$$\Rightarrow 1 - \frac{1}{x^2} = 0 \quad \Rightarrow \frac{1}{x^2} = 1$$

$$\Rightarrow x^2 = 1$$

$$\Rightarrow x = \pm 1$$

Now. When $x = 1$, then $f''(x) = \frac{2}{1^3}$
 $= 2 > 0$

So at $x = 1$, the given function will be minimum.

Hence the minimum value $= 1 + \frac{1}{1}$
 $= 2$ Ans

Again,

When $x = -1$, then $f''(x) = \frac{2}{-1^3}$
 $= -2 < 0$

So at $x = -1$ the given function will be maximum.

Hence the maximum value $= -1 - \frac{1}{1}$
 $= -2$ Ans

maximum value $= -2 < \text{minimum value} = 2$.

Hence the maximum value of $f(x)$ is

less than its minimum value. (shown)

③ prove that, the function $\sqrt{3} \sin x + 3 \cos x$ is maximum at $x = \frac{\pi}{6}$ and find its maximum value.

Given that

$$f(x) = \sqrt{3} \sin x + 3 \cos x$$

$$\therefore f'(x) = \sqrt{3} \cos x - 3 \sin x$$

$$\begin{aligned} \text{Also } f''(x) &= -\sqrt{3} \sin x - 3 \cos x \\ &= -(\sqrt{3} \sin x + 3 \cos x) \end{aligned}$$

$$f'(x) = 0$$

$$\Rightarrow \sqrt{3} \cos x - 3 \sin x = 0$$

$$\Rightarrow 3 \sin x = \sqrt{3} \cos x$$

$$\Rightarrow \frac{\sin x}{\cos x} = \frac{1}{\sqrt{3}}$$

$$\begin{aligned} \Rightarrow \tan x &= \tan \frac{\pi}{6} \\ x &= \frac{\pi}{6} \end{aligned}$$

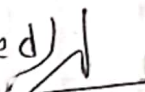
Now when $x = \frac{\pi}{6}$ then $f''(x) = -(\sqrt{3} \sin x + 3 \cos x)$

$$\begin{aligned} \therefore f''\left(\frac{\pi}{6}\right) &= -\left(\sqrt{3} \sin \frac{\pi}{6} + 3 \cos \frac{\pi}{6}\right) \\ &= -\left(\sqrt{3} \times \frac{1}{2} + 3 \times \frac{\sqrt{3}}{2}\right) \end{aligned}$$

$$= -\left(\sqrt{3}/2 + \frac{3\sqrt{3}}{2}\right)$$

$$= -\sqrt{3}/2 (1+3)$$

$$= -2\sqrt{3} < 0$$

So at $x = \pi/6$ the given function will be maximum. (proved) 

$$\text{then maximum value} = \sqrt{3} \times \sin \frac{\pi}{6} + 3 \cos \frac{\pi}{6}$$

$$= \sqrt{3} \times \frac{1}{2} + 3 \times \frac{\sqrt{3}}{2}$$

$$= 2\sqrt{3} \quad \text{proved Ans} \quad \text{✓}$$

④ Find max and min value of the following function,

$$1 + 2\sin x + 3\cos^2 x \quad (0 \leq x \leq \frac{\pi}{2})$$

$$\text{Let, } f(x) = 1 + 2\sin x + 3\cos^2 x$$

$$\therefore f'(x) = 2\cos x - 6\sin x \cos x$$

$$f''(x) = -2\sin x - 6\cos^2 x + 6\sin^2 x$$

$$f'(x) = 0$$

$$\Rightarrow 2 \cos x - 6 \sin x \cos x = 0$$

$$\Rightarrow 2 \cos x (1 - 3 \sin x) = 0$$

$$\Rightarrow \cos x = 0$$

$$\Rightarrow x = \cos^{-1} 0$$

$$= \pi/2 \quad \left[\because 0 \leq x \leq \frac{\pi}{2} \right)$$

Chap 3 Indefinite Integral

Type 1:

$$\int \frac{dx}{(ax+b)^n}$$

Shortcut

$$= \frac{1}{a(-n+1)(ax+b)^{n-1}} + C$$

Q1 Evaluate the integral $\int \frac{dx}{(3x+2)^5}$

Solution

$$\text{Let } I = \int \frac{dx}{(3x+2)^5}$$

$$\text{let } 3x+2 = z$$

$$\therefore 3dx = dz$$

$$\Rightarrow dx = \frac{1}{3} dz$$

$$= \int \frac{\frac{1}{3} dz}{z^5}$$

$$= \frac{1}{3} \int z^{-5} dz$$

$$= \frac{1}{3} \frac{z^{-5+1}}{-5+1} + C$$

$$= \frac{1}{12} \cdot z^{-4} + C$$

$$= -\frac{1}{12} \frac{1}{(3x+2)^4} + C$$

Type 2:

$$\int \frac{dx}{ax^2+bx+c}$$

ଦ୍ଵିଗତ 2x ଓ 1

x^2 ଏବଂ ଧନ common

ନିତେ ଏହା ଏବଂ

$(a+b)^2/(a-b)^2$ formula ଯେତେ 2(6-

formula ଗୋଟିଏ constant

ଅଣ/ଅଣିଏ ନାହିଁ ନିଶ୍ଚି variable

ବାହାରି ଗୋଟିଏ

1) Evaluate the integral $\int \frac{dx}{4x^2+8x+13}$

$$\text{Let, } I = \int \frac{dx}{4x^2+8x+13}$$

$$= \int \frac{dx}{4(x^2+2x+13/4)}$$

$$= 1/4 \int \frac{dx}{x^2+2x \cdot 1 + 1 + 13/4 - 1}$$

$$= 1/4 \int \frac{dx}{(x+1)^2 + (3/2)^2}$$

$$= 1/4 \cdot \frac{1}{3/2} \cdot \tan^{-1} \frac{x+1}{3/2} + C$$

$$\int \frac{du}{a^2+u^2} = \frac{1}{a} \tan^{-1} \frac{u}{a}$$

$$= \frac{1}{6} \tan^{-1} \frac{2(x+1)}{3} + C$$

$$\int \frac{dx}{x^2 + a^2} = \frac{1}{a} \tan^{-1} \frac{x}{a}$$

Formula

2) Evaluate the integral $\int \frac{dx}{x^2 - 6x + 5}$.

$$I = \int \frac{dx}{x^2 - 6x + 5}$$

$$= \int \frac{dx}{x^2 - 2 \cdot x \cdot 3 + 3^2 - 4} = \int \frac{dx}{(x-3)^2 - 4}$$

$$= \int \frac{dx}{(x-3)^2 - 2^2}$$

$$= \frac{1}{2 \cdot 2} \log \frac{x-3-2}{x-3+2}$$

$$= \frac{1}{4} \log \frac{x-5}{x-1}$$

$$\int \frac{dx}{x^2 - p^2} = \frac{1}{2p} \ln \left| \frac{x-p}{x+p} \right|$$

3) Evaluate the integral $\int \frac{dx}{4x^2 + 1}$

$$I = \int \frac{dx}{4x^2 + 1}$$

$$= \int \frac{dx}{4(x^2 + 1/4)} = \frac{1}{4} \int \frac{dx}{x^2 + (1/2)^2}$$

$$= \frac{1}{4} \cdot \frac{1}{1/2} \tan^{-1} \left| \frac{x}{1/2} \right| + C$$

$$= \frac{1}{2} \tan^{-1} 2x + C$$

4) Evaluate the integral $\int \frac{1}{4x^2 + 9} dx$

$$I = \int \frac{dx}{4x^2 + 9}$$

$$= \int \frac{dx}{4(x^2 + 9/4)}$$

$$= \frac{1}{4} \int \frac{dx}{x^2 + \left(\frac{3}{2}\right)^2}$$

$$= \frac{1}{4} \cdot \frac{1}{3/2} \tan^{-1} \frac{x}{3/2} + C$$

$$= \frac{1}{6} \tan^{-1} \frac{2x}{3} + C$$

5) Evaluate the integral $\int \frac{dx}{16x^2 - 9}$

$$I = \int \frac{dx}{16x^2 - 9}$$

$$= \int \frac{dx}{16(x^2 - 9/16)}$$

$$= \frac{1}{16} \cdot \int \frac{dx}{x^2 - (3/4)^2}$$

$$= \frac{1}{16} \cdot \frac{1}{2 \cdot 3/4} \log \frac{x - 3/4}{x + 3/4}$$

$$= \frac{1}{24} \cdot \log \frac{x - 3/4}{x + 3/4}$$

Type 3

$$\int \frac{f'(x)}{f(x)} dx = \ln |f(x)| + C$$

— ନିମ୍ନ function କେ diff. କରାଯାଇ ଡିମୋ function
— ଲାଗୁ ହେଉଛି ଇଣ୍ଟିଗ୍ରେସନ୍ ଲଗ୍ ଅଫ୍ ନିମ୍ନ
function.

1) Evaluate the integral $\int \frac{x}{1-x^2} dx$

$$\text{Let } I = \int \frac{x}{1-x^2} dx$$

$$= -\frac{1}{2} \int \frac{-2x}{1-x^2} dx$$

$$= -\frac{1}{2} \ln |1-x^2| + C$$

Type 4 :

$$\int \frac{px+q}{ax^2+bx+c} dx$$

1) Evaluate the integral $\int \frac{x dx}{4x^2+8x+13}$

$$I = \int \frac{x dx}{4x^2+8x+13}$$

$$= \int \frac{1/8(8x+8) - 1}{4x^2+8x+13} dx$$

$$= \int \frac{1/8(8x+8)}{4x^2+8x+13} - \int \frac{dx}{4x^2+8x+13}$$

$$= 1/8 \cdot \ln |4x^2+8x+13| - \int \frac{dx}{4(x^2+2x+13/4)}$$

$$= 1/8 \ln |4x^2+8x+13| - 1/4 \int \frac{dx}{(x+1)^2+(3/2)^2}$$

$$= 1/8 \ln |4x^2+8x+13| - 1/4 \cdot \frac{1}{3/2} \tan^{-1} \frac{x+1}{3/2}$$

$$= 1/8 \ln |4x^2+8x+13| - 1/6 \cdot \tan^{-1} \frac{2(x+1)}{3} + c$$



Method 5 :

$$\int \frac{px^2 + qx + r}{ax^2 + bx + c} dx$$

Q. Evaluate the integral $\int \frac{x^2}{x^2 - 4} dx$

Soln:

$$I = \int \frac{x^2}{x^2 - 4} dx$$

$$= \int \frac{(x^2 - 4) + 4}{(x^2 - 4)} dx$$

$$= \int \frac{x^2 - 4}{x^2 - 4} dx + \int \frac{4}{x^2 - 4} dx$$

$$= \int dx + 4 \int \frac{dx}{x^2 - 2^2}$$

$$= x + 4 \cdot \frac{1}{2 \cdot 2} \ln \left| \frac{x-2}{x+2} \right| + C$$

$$= x + \ln \left| \frac{x-2}{x+2} \right| + C$$

(Q) $\int \frac{dx}{x^2 - a^2} = \frac{1}{2a} \ln \left| \frac{x-a}{x+a} \right| + C$

Method 6: $\int \frac{px+q}{\sqrt{ax^2+bx+c}} dx$ Form, using $\int \frac{f'(x)}{\sqrt{f(x)}} dx = 2\sqrt{f(x)} + C$

Q. Evaluate the integral $\int \frac{x-2}{\sqrt{2x^2-8x+5}} dx$

Solution:

$$\text{Let, } I = \int \frac{x-2}{\sqrt{2x^2-8x+5}} dx$$

$$= \int \frac{\frac{1}{4}(4x-8)}{\sqrt{2x^2-8x+5}} dx$$

$$= \frac{1}{4} \int \frac{4x-8}{\sqrt{2x^2-8x+5}} dx$$

$$= \frac{1}{4} \cdot 2 \sqrt{2x^2-8x+5} + C$$

$$= \frac{1}{2} \sqrt{2x^2-8x+5} + C \quad (\text{Ans.})$$

Method 7:

$$\int \frac{dx}{(ax+b)\sqrt{cx+d}} \quad (a \neq 0, c \neq 0) \text{ then let, } cx+d = z^2$$

Q. Evaluate the integral, $\int \frac{dx}{(2+x)\sqrt{1+x}}$

Solution:

$$\text{Let, } I = \int \frac{dx}{(2+x)\sqrt{1+x}}$$

$$= \int \frac{2z dz}{(2+z^2-1) \cdot z}$$

$$\text{Let, } 1+x = z^2$$

$$dx = 2z dz$$

$$= 2 \int \frac{dz}{1+z^2}$$

$$= 2 \tan^{-1} z + c$$

$$= 2 \tan^{-1} \sqrt{1+x} + c$$

Method 8: $\int \frac{dx}{(px+q)\sqrt{ax^2+bx+c}}$ form, let $px+q = \frac{1}{z}$

Q. Evaluate the Integral $\int \frac{dx}{(2x+3)\sqrt{x^2+3x+2}}$

Solution;

Let, $I = \int \frac{dx}{(2x+3)\sqrt{x^2+3x+2}}$

let, $2x+3 = \frac{1}{z}$
 $2dx = \frac{-1}{z^2} dz$
 or, $dx = -\frac{1}{2z^2} dz$

$= \int \frac{-\frac{1}{2z^2} dz}{\frac{1}{z} \sqrt{\frac{1}{4}\left(\frac{1}{z}-3\right)^2 + 3 \cdot \frac{1}{2}\left(\frac{1}{z}-3\right) + 2}}$

$= -\frac{1}{2} \int \frac{\frac{1}{z} dz}{\sqrt{\frac{1}{4}z^2 - \frac{6}{4z} + \frac{9}{4} + \frac{3}{2z} - \frac{9}{2} + 2}}$

$= -\frac{1}{2} \int \frac{\frac{1}{z} dz}{\sqrt{\frac{1-6z+9z^2+6z-18z^2+8z^2}{4z^2}}}$

$= -\frac{1}{2} \int \frac{\frac{1}{z} \cdot z dz}{\sqrt{1-z^2}}$

$= -\int \frac{dz}{\sqrt{1-z^2}} = -\sin^{-1} z + C$

$= -\sin^{-1}\left(\frac{1}{2x+3}\right) + C$

Type 9

$$\text{using } \int uv \, dx = u \int v \, dx - \int \left\{ \frac{d}{dx} (u) \right.$$

$$\left. \int v \, dx \right\} dx$$

① Evaluate the integral $\int x \tan^{-1} x \, dx$

Sol:

$$\text{Let } I = \int x \tan^{-1} x \, dx$$

যাতে Integration এর
কড়িট গঠন করা হবে

$$= \tan^{-1} x \int x \, dx - \int \left\{ \frac{d}{dx} (\tan^{-1} x) \int x \, dx \right\} dx$$

$$= \tan^{-1} x \cdot \frac{x^2}{2} - \int \frac{1}{1+x^2} \cdot \frac{x^2}{2} \, dx$$

$$\begin{aligned} u &= \tan^{-1} x \\ v &= x \end{aligned}$$

$$= \frac{1}{2} x^2 \tan^{-1} x - \frac{1}{2} \int \frac{x^2}{1+x^2} \, dx$$

$$= \frac{1}{2} x^2 \tan^{-1} x - \frac{1}{2} \int \frac{(1+x^2)-1}{1+x^2} \, dx$$

$$= \frac{1}{2} x^2 \tan^{-1} x - \frac{1}{2} \int \frac{1+x^2}{1+x^2} \, dx + \frac{1}{2} \int \frac{1 \, dx}{1+x^2}$$

$$= \frac{1}{2} x^2 \tan^{-1} x - \frac{1}{2} x + \frac{1}{2} \tan^{-1} x + C.$$

Type 10

using $\int e^x \{ f(x) + f'(x) \} dx = e^x f(x) + C$

① Evaluate the integral

$$\int e^x (\sin x - \cos x) dx.$$

$f(x)$ to differentiation or
 $f'(x)$ or integration or both format
or any

Sol: Let, $I = \int e^x (\sin x - \cos x) dx$

Again let,

$$f(x) = -\cos x$$

$$f'(x) = \sin x$$

$-\cos x$ to differentiation
or $\sin x$ or integration
or $f(x) = -\cos x, f'(x) = \sin x$

therefore

$$I = \int e^x \{ f(x) + f'(x) \} dx$$

$$= e^x f(x) + C$$

$$= -e^x \cos x + C$$

Group B

Chapter 4 : Geometrical interpretation.

Formula $\int_a^b f(x) dx = \lim_{h \rightarrow 0} \sum_{r=0}^{n-1} h \cdot f(a + r h)$

where $rh = b - a$

① Evaluate the integral $\int_0^1 x dx$ by geometrical interpretation.

Sol:

we know that,

$$\int_a^b f(x) dx = \lim_{h \rightarrow 0} \sum_{r=0}^{n-1} h \cdot f(a + r h) \quad \text{--- ①}$$

according to question here,

$$a = 0, b = 1$$

also,

$$f(x) = x$$

$$\therefore f(a + r h) = a + r h$$

$$= 0 + r h$$

$$= r h$$

So eqn ① becomes,

$$\int_0^1 x dx = \lim_{h \rightarrow 0} \sum_{n=0}^{n-1} h \cdot \pi h$$

$$= \lim_{h \rightarrow 0} h^2 \sum_{n=0}^{n-1} \pi$$

$$= \lim_{h \rightarrow 0} h^2 \sum_{n=0+1}^{n-1+1} \pi$$

$$= \lim_{h \rightarrow 0} h^2 (1+2+3+\dots+n)$$

$$= \lim_{h \rightarrow 0} h^2 \frac{n(n+1)}{2}$$

$$= \lim_{h \rightarrow 0} \frac{nh(nh+h)}{2}$$

$$= \lim_{h \rightarrow 0} \frac{1(1+h)}{2}$$

$$= 1/2$$

$$\begin{aligned} \because nh &= b-a \\ &= 1-0=1 \end{aligned}$$

(11) Using reduction formula prove that,

$$\int \sin^n x \, dx = -\frac{1}{n} \sin^{n-1} x \cos x + \frac{n-1}{n} \int \sin^{n-2} x \, dx$$

Solution:

$$\text{Let } I = \int \sin^n x \, dx \quad \text{--- (1)}$$

$$= \sin^{n-1} x \sin x \, dx$$

$$= \sin^{n-1} x \int \sin x - \int \left\{ \frac{d}{dx} (\sin^{n-1} x) \int \sin x \, dx \right\} dx$$

$$= \sin^{n-1} x (-\cos x) - \int (n-1) \sin^{n-2} x \cos x (-\cos x) dx$$

$$= -\sin^{n-1} x \cos x + (n-1) \int \sin^{n-2} x \cos^2 x \, dx$$

$$= -\sin^{n-1} x \cos x + (n-1) \int \sin^{n-2} x (1 - \sin^2 x) dx$$

$$= -\sin^{n-1} x \cos x + (n-1) \int \sin^{n-2} x \, dx - (n-1) \int \sin^n x \, dx$$

$$= -\sin^{n-1} x \cos x + (n-1) \int \sin^{n-2} x \, dx - (n-1) I \quad \text{by (1)}$$

$$\Rightarrow I + (n-1)I = -\sin^{n-1} x \cos x + (n-1) \int \sin^{n-2} x \, dx$$

$$\Rightarrow I = -\frac{1}{n} \sin^{n-1} x \cos x + \frac{n-1}{n} \int \sin^{n-2} x \, dx$$

Hence

$$\int \sin^n x \, dx = -\frac{1}{n} \sin^{n-1} x \cos x +$$

$$\frac{n-1}{n} \int \sin^{n-2} x \, dx. \underline{\quad}$$

(P2) Using reduction formula prove that,

$$\int \sin^2 x \, dx = -1/2 \sin x \cos x + 1/2 x + c$$

solution: we know by the reduction formula

$$\int \sin^n x \, dx = -1/n \sin^{n-1} x \cos x + \frac{n-1}{n} \int \sin^{n-2} x \, dx$$

now putting $n=2$, we get.

$$\int \sin^2 x \, dx = -1/2 \sin x \cos x + 1/2 \int dx$$

$$= -1/2 \sin x \cos x + 1/2 x + c$$

⑦③ ⑤ If $I_n = \int_0^{\pi/4} \tan^n \theta \, d\theta$ and $n > 1$ then using reduction formula show that, $n(I_{n-1} + I_{n+1}) = 1$

Solution

Given that, $I_n = \int_0^{\pi/4} \tan^n \theta \, d\theta$ — (i)

$\therefore I_{n-1} = \int_0^{\pi/4} \tan^{n-1} \theta \, d\theta$ — (ii)

Also $I_{n+1} = \int_0^{\pi/4} \tan^{n+1} \theta \, d\theta$
 $= \int_0^{\pi/4} \tan^{n-1} \theta \cdot \tan^2 \theta \, d\theta$

$$= \int_0^{\pi/4} \tan^{n-1} \theta (\sec^2 \theta - 1) d\theta$$

$$= \int_0^{\pi/4} \tan^{n-1} \theta \sec^2 \theta d\theta - \int_0^{\pi/4} \tan^{n-1} \theta d\theta$$

$$= \int_0^{\pi/4} \tan^{n-1} \theta \sec^2 \theta d\theta - I_{n-1} \quad [\text{by (1)}]$$

$$\Rightarrow I_{n-1} + I_{n+1} = \int_0^{\pi/4} \tan^{n-1} \theta \sec^2 \theta d\theta$$

$$= \int_0^1 z^{n-1} dz$$

$$= \left[\frac{z^{n-1+1}}{n-1+1} \right]_0^1$$

$$= \frac{1}{n} [z^n]_0^1$$

$$= \frac{1}{n}$$

Let
 $\tan \theta = z$
 $\therefore \sec^2 \theta = dz$
 Also,

θ	0	$\pi/4$
z	0	1

$$\text{Hence } n(I_{n-1} + I_{n+1}) = 1$$

Chapter

Definite Integral

Formula

$$(i) \int_a^b f(x) dx = \int_a^b f(z) dz$$

$$(ii) \int_a^b f(x) dx = - \int_b^a f(x) dx$$

$$(iii) \int_a^b f(x) dx = \int_a^c f(x) dx + \int_c^b f(x) dx, \quad a < c < b$$

$$(iv) \int_0^a f(x) dx = \int_0^a f(a-x) dx$$

$$(v) \int_{-a}^a f(x) dx = 2 \int_0^a f(x) dx, \text{ when } f(x) \text{ is even}$$

$$(vi) \int_{-a}^a f(x) dx = 0, \text{ when } f(x) \text{ is odd.}$$

1 Show that, $\int_0^{\pi/4} \frac{\sin x}{\sin x + \cos x} dx = \pi/4$

Solution

$$\text{Let } I = \int_0^{\pi/2} \frac{\sin x}{\sin x + \cos x} dx \quad \text{--- (1)}$$

$$= \int_0^{\pi/2} \frac{\sin(\pi/2 - x)}{\sin(\pi/2 - x) + \cos(\pi/2 - x)} dx \quad \text{--- (by (iv) formula)}$$

$$\Rightarrow I = \int_0^{\pi/2} \frac{\cos x}{\cos x + \sin x} dx \quad \text{--- (2)}$$

Now adding, ① & ② we get,

$$2I = \int_0^{\pi/2} \frac{\sin x + \cos x}{\sin x + \cos x} dx$$

$$= \int_0^{\pi/2} dx = [x]_0^{\pi/2}$$

$$2I = \pi/2$$

$$I = \pi/4$$

Hence $\int_0^{\pi/2} \frac{\sin x}{\sin x + \cos x} dx = \pi/4$ 

Chapter 8: Successive Integral

① Evaluate the triple integral $\int_0^{3y} \int_2^3 \int_1^{yz} (2x + y + z) dx dy dz$

Solution

$$\begin{aligned} \text{Let } I &= \int_0^{3y} \int_2^3 \int_1^{yz} (2x + y + z) dx dy dz \\ &= \int_2^3 \int_0^{3y} \int_1^{yz} (2x + y + z) dx dz dy \\ &= \int_2^3 \int_0^{3y} \left[2 \cdot \frac{x^2}{2} + y \cdot x + z \cdot x \right]_{x=1}^{yz} dz dy \\ &= \int_2^3 \int_0^{3y} (y^2 z^2 + y^2 z + z^2 y - 1 - y - z) dz dy \\ &= \int_2^3 \left[y^2 \cdot \frac{z^3}{3} + y^2 \frac{z^2}{2} + y \cdot \frac{z^3}{3} - z - y \cdot z - \frac{z^2}{2} \right]_{z=0}^{3y} dy \\ &= \int_2^3 \left(9y^5 + \frac{9}{2} y^4 + 9y^4 - 3y - 3y^2 - \frac{9}{2} y^2 \right) dy \\ &= \left[9 \cdot \frac{y^6}{6} + \frac{9}{2} \frac{y^5}{5} + \frac{y^5}{5} 9 - 3 \frac{y^2}{2} - 3 \frac{y^3}{3} - \frac{9}{2} \frac{y^3}{3} \right]_{y=2}^3 \\ &= \left(\frac{3}{2} \cdot 3^6 + \frac{9}{10} 3^5 + \frac{9}{5} \cdot 3^5 - \frac{3}{2} 3^2 - 3^3 - \frac{3}{2} 3^3 \right) - \\ &\quad \left(\frac{3}{2} 2^6 + \frac{9}{10} 2^5 + \frac{9}{5} 2^5 - \frac{3}{2} 2^2 - 2^3 - \frac{3}{2} 2^3 \right) \end{aligned}$$

$$= \left(\frac{5 \times 3^7 + 9 \times 3^5 + 18 \times 3^5 - 3^3 \times 5 - 10 \times 3^3 - 5 \times 3^4}{10} \right) - \left(\frac{15 \times 2^6 + 9 \times 2^5 + 18 \times 2^5 - 15 \times 2^2 - 10 \times 2^3 - 15 \times 2^3}{10} \right)$$

$$= \frac{8343}{5} - \frac{782}{5}$$

$$= \frac{8343 - 782}{5} = \frac{7561}{5} = 1512.2 \quad \text{Ans}$$

Chapter 8 : The Gamma and Beta function

Gamma function: The integral $\int_0^{\infty} x^{n-1} \cdot e^{-x} dx$ is called the second Eulerian integral. This function defined for positive values of n ($n > 0$) is known as gamma function and denoted by Γn

$$\Gamma n = \int_0^{\infty} x^{n-1} e^{-x} dx$$

Beta function: The integral $\int_0^1 x^{m-1} (1-x)^{n-1} dx$ is called the first Eulerian integral. This function defined for positive values of m and n ($m > 0, n > 0$) is known as Beta function, and denoted by $\beta(m, n)$.

Formulae:

2. $\Gamma 1 = 1$

3. $\Gamma(n+1) = n \Gamma n$

4. $\Gamma n = (n-1)! = (n-1) \cdot (n-2) \cdot \dots \cdot 1$

5. $\beta(m, n) = \beta(n, m)$

9. $\Gamma 1/2 = \sqrt{\pi}$

10. $\int_0^{\pi/2} \sin^m x \cdot \cos^n x dx = \frac{\Gamma(\frac{m+1}{2}) \Gamma(\frac{n+1}{2})}{2 \Gamma(\frac{m+n+2}{2})} = \beta(m, n)$

6. $\beta(m, n) = 2 \int_0^{\pi/2} \sin^{2m-1} \theta \cos^{2n-1} \theta d\theta$

7. $\beta(m, n) = \frac{\Gamma(m-1) \Gamma(n-1)}{\Gamma(m+n-1)}$

8. $\beta(m, n) = \frac{\Gamma m \Gamma n}{\Gamma(m+n)}$

$$11. \int_0^{\pi/2} \sin^m x \, dx = \frac{1}{2} \sqrt{\pi} \frac{\Gamma(\frac{m+1}{2})}{\Gamma(\frac{m+2}{2})}$$

$$12. \int_0^{\pi/2} \cos^n x \, dx = \frac{1}{2} \sqrt{\pi} \frac{\Gamma(\frac{n+1}{2})}{\Gamma(\frac{n+2}{2})}$$

(Q.1) prove that $\beta(m, n) = 2 \int_0^{\pi/2} \sin^{2m-1} \theta \cdot \cos^{2n-1} \theta \, d\theta$

Sol.

We know,
$$\beta(m, n) = \int_0^1 x^{m-1} (1-x)^{n-1} \, dx$$

Let, $x = \sin^2 \theta$

$\therefore dx = 2 \sin \theta \cos \theta \, d\theta$

Also,

x	0	1
θ	0	$\pi/2$

Therefore,
$$\beta(m, n) = \int_0^{\pi/2} (\sin^2 \theta)^{m-1} (1 - \sin^2 \theta)^{n-1} \cdot 2 \sin \theta \cos \theta \, d\theta$$

$$= 2 \int_0^{\pi/2} \sin^{2m-2} \theta \cos^{2n-2} \theta \cdot \sin \theta \cos \theta \, d\theta$$

$$= 2 \int_0^{\pi/2} \sin^{2m-1} \theta \cos^{2n-1} \theta \, d\theta$$

Hence
$$p(m, n) = 2 \int_0^{\pi/2} \sin^{2m-1} \theta \cos^{2n-1} \theta d\theta. \quad \underline{\underline{\text{proved}}}$$

Q2

prove that $\Gamma_{1/2} = \sqrt{\pi}$

sol

We know that,

$$\beta(m, n) = 2 \int_0^{\pi/2} \sin^{2m-1} \theta \cos^{2n-1} \theta d\theta$$

putting, $m = n = 1/2$ we get,

$$\begin{aligned} p(1/2, 1/2) &= 2 \int_0^{\pi/2} d\theta \\ &= 2 [\theta]_0^{\pi/2} \\ &= 2 \pi/2 \\ &= \pi \quad \text{--- ①} \end{aligned}$$

Again we know that,

$$\beta(m, n) = \frac{\Gamma_m \Gamma_n}{\Gamma(m+n)}$$

putting, $m = n = 1/2$ we set

$$\begin{aligned} B(1/2, 1/2) &= \frac{\Gamma_{1/2} \Gamma_{1/2}}{\Gamma(1/2 + 1/2)} \\ &= \frac{(\Gamma_{1/2})^2}{\Gamma_1} \end{aligned}$$

$$= \Gamma_{1/2}^2 \text{ --- (2) } [\because \Gamma_1 = 1]$$

Now from (1) & (2) we have,

$$(\Gamma_{1/2})^2 = \pi$$

$$\text{Hence } \Gamma_{1/2} = \sqrt{\pi} \quad (\text{proved})$$

Chapter 9

Finding Arc length of a curve,

Formula:

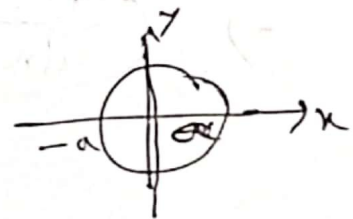
$$1. L = \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx \quad \text{about } x \text{ axis}$$

$$2. L = \int_a^b \sqrt{1 + \left(\frac{dx}{dy}\right)^2} dy \quad \text{about } y \text{ axis}$$

(a1) Find the circumference of a circle
 $x^2 + y^2 = a^2$ where a is the radius of the
 circle.

Sol Given that, eqn of circle,

$$x^2 + y^2 = a^2 \quad \text{--- (1)}$$



Whose center is at $(0,0)$ and
 radius a .

Now from (1) we get,

$$y = \pm \sqrt{a^2 - x^2}$$

Which means that, the upper semi circle is $y = \sqrt{a^2 - x^2}$ and the lower ~~case~~ semi-circle is $y = -\sqrt{a^2 - x^2}$.

Now we will find the upper semi-circle's circumference length, i.e.,

$$y = \sqrt{a^2 - x^2}$$

$$\therefore \frac{dy}{dx} = \frac{1}{2\sqrt{a^2 - x^2}} (-2x)$$

$$= \frac{-x}{\sqrt{a^2 - x^2}}$$

$$\Rightarrow \left(\frac{dy}{dx}\right)^2 = \frac{x^2}{\cancel{a^2 - x^2} a^2 - x^2}$$

$$\Rightarrow 1 + \left(\frac{dy}{dx}\right)^2 = 1 + \frac{x^2}{a^2 - x^2}$$

$$= \frac{a^2 - x^2 + x^2}{a^2 - x^2}$$

$$= \frac{a^2}{a^2 - x^2}$$

$$\therefore \sqrt{1 + \left(\frac{dy}{dx}\right)^2} = \sqrt{\frac{a^2}{a^2 - x^2}}$$

$$= \frac{a}{\sqrt{a^2 - x^2}}$$

Hence the length of the circumference of upper semicircle, about x axis,

$$L = \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$

$$= \int_{-a}^a \frac{a}{\sqrt{a^2 - x^2}} dx \quad \left\{ \begin{array}{l} \text{lower limit } a = -a \\ \text{upper } \quad \quad b = a \end{array} \right.$$

$$= a \int_{-a}^a \frac{dx}{\sqrt{a^2 - x^2}} = a \left[\sin^{-1} \frac{x}{a} \right]_{-a}^a$$

$$= a \left[\sin^{-1} \frac{a}{a} - \sin^{-1} \left(\frac{-a}{a} \right) \right]$$

$$= a (\sin^{-1} 1 + \sin^{-1} 1)$$

$$= a (\pi/2 + \pi/2)$$

$$= a \frac{2\pi}{2}$$

$$= a\pi$$

Therefore the circumference of the given

circle

$$= 2 \times a \pi$$

$$= 2 \pi a.$$

#

Finding the Areas of surface of revolution.

formulas:

$$1. A = \int_a^b 2\pi y \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx \longrightarrow \text{about } x \text{ axis}$$

$$2. A = \int_a^b 2\pi x \sqrt{1 + \left(\frac{dx}{dy}\right)^2} dy, \text{ about } y \text{ axis}$$

Q1 Find the area of the surface of:

revolution obtained by revolving the graph

of $y = f(x) = 2x$ from $x = -3$ to $x = -1$

about the x axis.

Sol

Given that,

$$y = f(x) = 2x$$

$$\therefore \frac{dy}{dx} = 2$$

$$\Rightarrow \left(\frac{dy}{dx}\right)^2 = 4$$

$$\Rightarrow 1 + \left(\frac{dy}{dx}\right)^2 = 1 + 4 = 5$$

Now the required area of the graph
of $y = 2x$ from $x = -3$ to $x = -1$
about the x axis.

$$A = \int_a^b 2\pi y \cdot \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$

$$= \int_{-3}^{-1} 2\pi \cdot 2x \sqrt{5} dx \quad \begin{array}{l} \text{lower limit } a = -3 \\ \text{upper } b = -1 \end{array}$$

$$= 4\pi\sqrt{5} \int_{-3}^{-1} x dx$$

$$= 4\pi\sqrt{5} \left[\frac{x^2}{2} \right]_{-3}^{-1}$$

$$= \frac{4\pi\sqrt{5}}{2} \{ (-1)^2 - (-3)^2 \}$$

$$= -16\pi\sqrt{5} \quad \underline{\underline{A}}$$

(71) finding the volume of a solid of revolution.

formulas:

1. $V = \int_a^b \pi y^2 dx$ about x axis

2. $V = \int_a^b \pi x^2 dy$ about y axis

(Q1) finding the volume of the solid of revolution generated by rotating the curve $y = x^3$ between $y=0$ and $y=4$ about the y axis.

Sol

Given that

$$y = x^3$$

$$\Rightarrow x = y^{1/3}$$

Now the required value of the given curve between $y=0$ and

$y=4$ about y axis.

$$V = \int_a^b \pi r^2 dy$$

$$= \int_0^4 \pi (y^{1/3})^2 dy$$

low. limit $a=0$
upper lin $b=4$

$$= \pi \int_0^4 y^{2/3} dy$$

$$= \pi \left[\frac{y^{2/3+1}}{\frac{2}{3}+1} \right]_0^4$$

$$= \pi \frac{3}{5} \left[y^{5/3} \right]_0^4$$

$$= \frac{3}{5} \pi 4^{5/3}$$

